

Parent Guide



@twinklparents

We're excited to share this activity with you. If you are interested in finding more engaging, fun and interesting activities for you and your children, then check out these links to different areas of the [Twinkl Parents](#) website.

games



crafts



puzzles



experiments



word searches



What is this resource and how do I use it?

Support your year 3 child as they develop their understanding of 3D shapes with this fun Properties of 3D Shapes activity, which includes a variety of tasks for your child to complete as well as a quick guide. All you need to do is download, print off and get learning!

What skills does this practise?

Properties of Shapes

Geometry

3D Shapes

Further Activity Ideas and Suggestions

This [Parent Guide to Year 3 Maths](#) outlines everything your child should know by the end of the year. You can use these fun shapes [activity mats](#) to recap what they learnt in key stage 1 so that you know they are confident before you start this activity.

Parents Blog



Twinkl Kids' TV



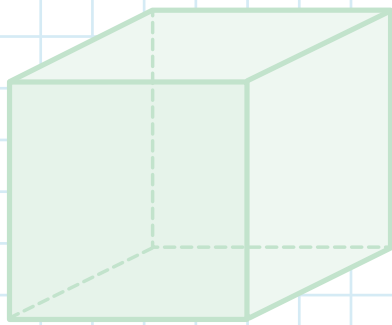
Homework Help



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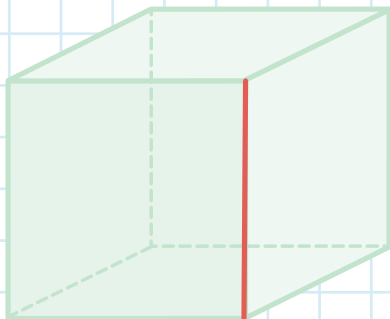
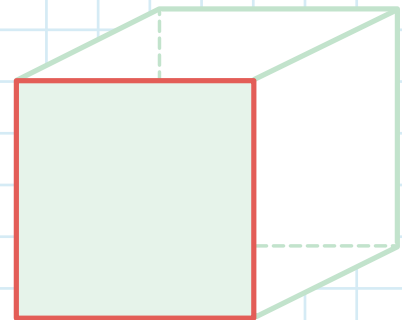
Parents
Hub

Properties of 3D Shapes



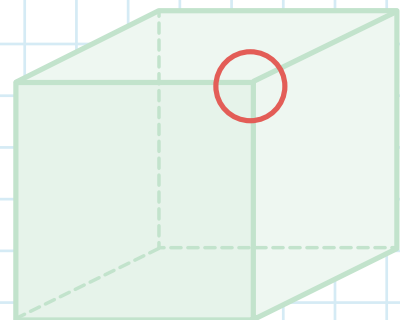
This 3D shape is a **cube**.

The 2D shape shown is one of the cube's **faces**. A cube has **6 faces** in total. They are all **squares**.

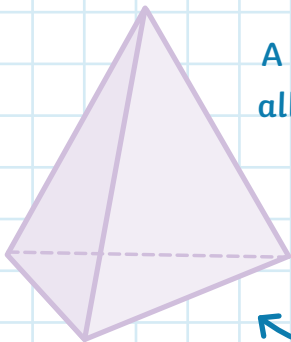


Where two faces meet is called an **edge**.
A cube has **12 edges** in total.

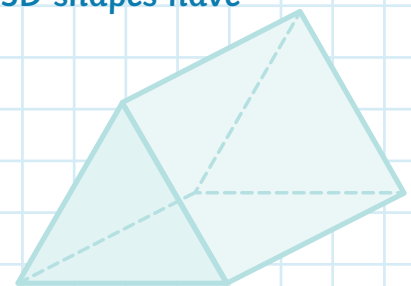
Where the edges meet is called a **vertex**.
A cube has **8 vertices** in total.



While a cube is composed of 6 square faces, many other 3D shapes have different 2D shapes as their faces.



A tetrahedron is made up of 4 faces, all of which are equilateral triangles (if it is a regular tetrahedron).



A triangular prism is made up of 5 faces - 2 triangles and 3 rectangles (these could be square rectangles or oblong rectangles).

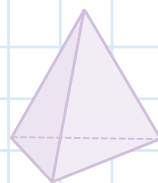
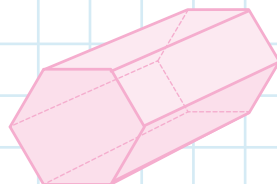
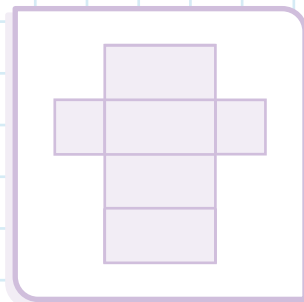
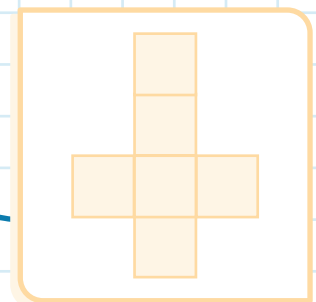
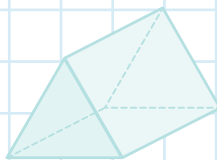
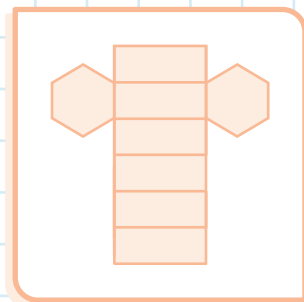
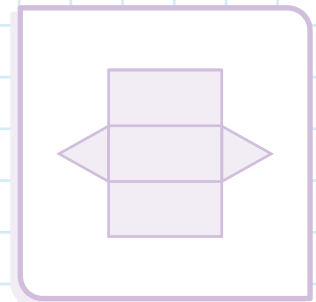
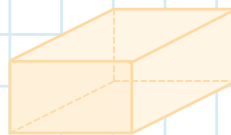
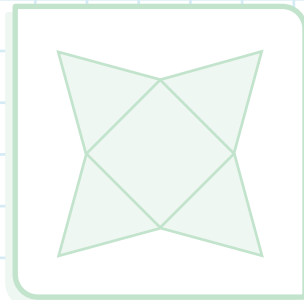
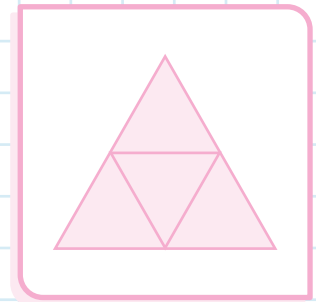
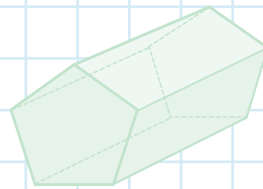
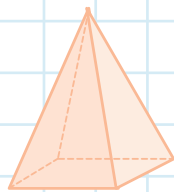
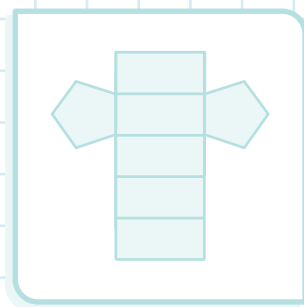
3D Shape Nets

3D shapes are often shown as 2D representations - as **nets**.

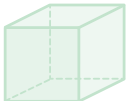
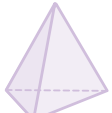
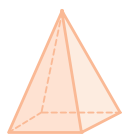
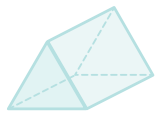
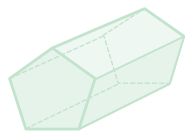
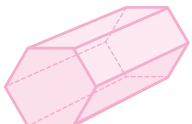
It might help to imagine that a net is what a 3D shape might look like before it is folded up and glued together.

Nets are useful for helping you to see which different 2D shapes make up a 3D shape.

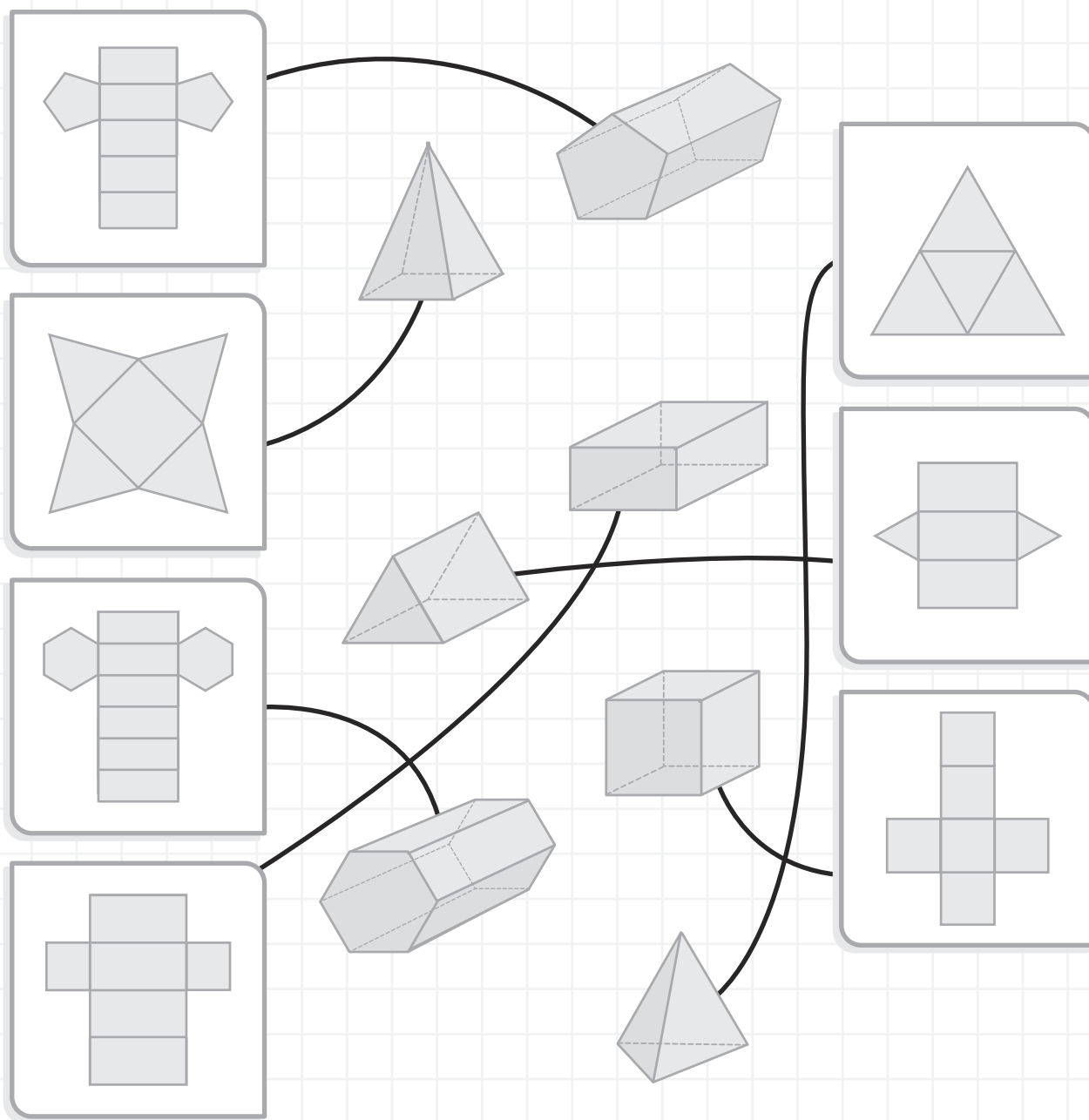
Can you match the nets below to the 3D shapes? The first one has been done for you.



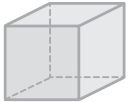
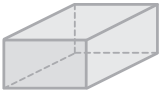
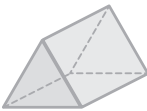
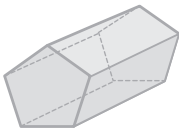
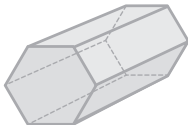
Properties of 3D Shapes

	Shape Name	Number of Faces	Types of Faces	Number of Edges	Number of Vertices
	cube	6	6 squares	12	8
					
					
					
					
					
					

3D Shape Nets - Answers



Properties of 3D Shapes – Answers

	Shape Name	Number of Faces	Types of Faces	Number of Edges	Number of Vertices
	cube	6	6 squares	12	8
	cuboid	6	6 rectangles	12	8
	tetrahedron	4	4 equilateral triangles	6	4
	square-based pyramid	5	1 square 4 equilateral triangles	8	5
	triangular prism	5	3 rectangles 2 triangles	9	6
	pentagonal prism	7	2 pentagons 5 rectangles	15	10
	hexagonal prism	8	2 hexagons 6 rectangles	18	12

We hope you find the information on our website and resources useful. As far as possible, the contents of this resource are reflective of current professional research. However, please be aware that every child is different and information can quickly become out of date. The information given here is intended for general guidance purposes only and may not apply to your specific situation.



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